WORKED SOLUTION

STEP I 2017, Question 13

Probability and Statistics • Recurrence Relations for Probabilities

STEP I 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q13

The key idea

Look at each slice and ask how it is used: it can be toast, the first slice of a sandwich, or the second slice of a sandwich. Exactly one of these happens, and the slice before it decides which kinds are possible. If slice $r-1$ finishes a snack, slice r starts a new one; if it doesn't, slice r must be the second half of a sandwich. That gives a first-order recurrence, which we solve by the usual trick of subtracting off the fixed point.

Solution

Setting up

Number the slices $1, 2, \ldots, n$ in the order they are used. For each slice r exactly one of the following happens:

- T_r: slice r is used for toast;
- A_r: slice r is the first of the two slices in a sandwich;
- S_r: slice r is the second of the two slices in a sandwich.

So $t_r = \mathrm{P}(T_r)$, $s_r = \mathrm{P}(S_r)$ and $\mathrm{P}(A_r) = 1 - s_r - t_r$. Say that slice r *ends a snack* if T_r or S_r happens; this has probability $s_r + t_r$.

The first slice starts the first snack, so it cannot be the second slice of a sandwich:

$$s_1 = 0.$$

(Also $t_1 = p$ if $n \geq 2$, and $t_1 = 1$ if $n = 1$.)

The two equations

The equation for t_r. Slice r starts a new snack exactly when slice $r-1$ ends a snack, which has probability $s_{r-1} + t_{r-1}$. Given that slice r starts a snack, slices $r, r+1, \ldots, n$ are left, and when $r \leq n-1$ that is at least two slices, so I choose toast with probability

p, whatever happened earlier. If slice $r-1$ does not end a snack, slice r completes a sandwich and is not toast. Hence

$$t_r = (s_{r-1} + t_{r-1})\, p \qquad (2 \le r \le n-1).$$

This fails for $r = n$ because a snack starting at the last slice is toast for certain.

The equation for s_r. Slice r is the second slice of a sandwich exactly when slice $r-1$ is the first slice of one, that is, when slice $r-1$ does *not* end a snack. So

$$s_r = 1 - (s_{r-1} + t_{r-1}) \qquad (2 \le r \le n).$$

No condition on the number of slices left is needed here, so this holds right up to $r = n$.

The recurrence for s_r

The second equation, applied with r replaced by $r+1$, gives $s_r + t_r = 1 - s_{r+1}$, so the first equation can be written as

$$t_r = p\,(1 - s_r) \qquad (2 \le r \le n-1).$$

This is also true for $r = 1$ when $n \ge 2$, as $t_1 = p$ and $s_1 = 0$. Substituting $t_{r-1} = p(1 - s_{r-1})$ into the equation for s_r, valid when $1 \le r-1 \le n-1$:

$$\begin{aligned} s_r &= 1 - s_{r-1} - p(1 - s_{r-1}) \\ &= (1-p)(1 - s_{r-1}) \\ &= q\,(1 - s_{r-1}), \end{aligned}$$

for $2 \le r \le n$, and in particular for $2 \le r \le n-1$, as required.

Solving the recurrence

> **Ask yourself** How do we solve $s_r = q - q s_{r-1}$? What constant sequence satisfies it?

A constant solution $s_r = c$ needs $c = q - qc$, so $c = \dfrac{q}{1+q}$. Subtracting $c = q - qc$ from the recurrence,

$$s_r - \frac{q}{1+q} = -q\left(s_{r-1} - \frac{q}{1+q}\right),$$

so $s_r - \frac{q}{1+q}$ is multiplied by $-q$ at each step. Starting from $s_1 - \frac{q}{1+q} = -\frac{q}{1+q}$,

$$s_r - \frac{q}{1+q} = (-q)^{r-1}\left(-\frac{q}{1+q}\right) = \frac{(-q)^r}{1+q},$$

which gives

$$s_r = \frac{q + (-q)^r}{1+q} \qquad (1 \le r \le n-1).$$

(Since the recurrence holds up to $r = n$, the same formula actually holds for $r = n$ too.)

For t_r, we use $t_r = p(1 - s_r)$ and

$$1 - s_r = \frac{1 + q - q - (-q)^r}{1 + q} = \frac{1 - (-q)^r}{1 + q},$$

so

$$t_r = \frac{p\big(1 - (-q)^r\big)}{1 + q} \qquad (1 \le r \le n - 1).$$

The last slice

As noted above, the equation for s_r and the recurrence $s_r = q(1 - s_{r-1})$ hold for $r = n$ as well, so the formula continues:

$$s_n = \frac{q + (-q)^n}{1 + q}.$$

The last slice cannot be the first slice of a sandwich, as there is nothing after it. So it is either toast or the second slice of a sandwich, and $t_n = 1 - s_n$:

$$t_n = \frac{1 - (-q)^n}{1 + q}.$$

Compared with t_r for $r < n$, the factor p has gone: once a snack starts on the last slice, toast is certain. (These also hold when $n = 1$: they give $s_1 = 0$ and $t_1 = 1$.)

Checks

- Take $n = 3$ and $p = q = \frac{1}{2}$. The possible orders are toast–toast–toast (probability $\frac{1}{4}$), toast then a sandwich (probability $\frac{1}{4}$), and a sandwich then toast (probability $\frac{1}{2}$). So $s_3 = \frac{1}{4}$ and $t_3 = \frac{3}{4}$. The formulae give $s_3 = \dfrac{\frac{1}{2} - \frac{1}{8}}{\frac{3}{2}} = \frac{1}{4}$ and $t_3 = \dfrac{1 + \frac{1}{8}}{\frac{3}{2}} = \frac{3}{4}$.
- With $q = 0$ (always toast) we get $s_r = 0$ and $t_r = 1$ for every r, as we should.
- As r grows, $s_r \to \frac{q}{1+q}$. This makes sense: in the long run a fraction q of snacks are sandwiches, each with one "second slice", and a snack uses $1 + q$ slices on average, so a proportion $\frac{q}{1+q}$ of slices are second slices.
- We confirmed all the formulae by computer, using exact enumeration of every snack sequence for $n \le 12$ and several values of p.

Common slips

- Using $t_n = p(s_{n-1} + t_{n-1})$: when a snack starts on the last slice it is toast with probability 1, not p.

- Thinking the events "slice $r-1$ is toast" and "slice r is toast" are independent. They are not: what matters is whether slice r starts a snack, and that depends on the history.
- Losing the sign when solving the recurrence: the ratio is $-q$, so the answers alternate about $\frac{q}{1+q}$.

Beyond STEP

Whether slice r starts a new snack forms a simple *renewal process*: the snack lengths are independent, each 1 or 2, and the limit $\frac{q}{1+q}$ is an instance of the renewal theorem (the long-run rate of renewals is one over the mean gap). Equivalently, the type of each slice is a Markov chain, and $(-q)^r$ shows how fast it forgets its starting state. Grimmett and Welsh, *Probability: An Introduction*, covers both Markov chains and recurrences of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.